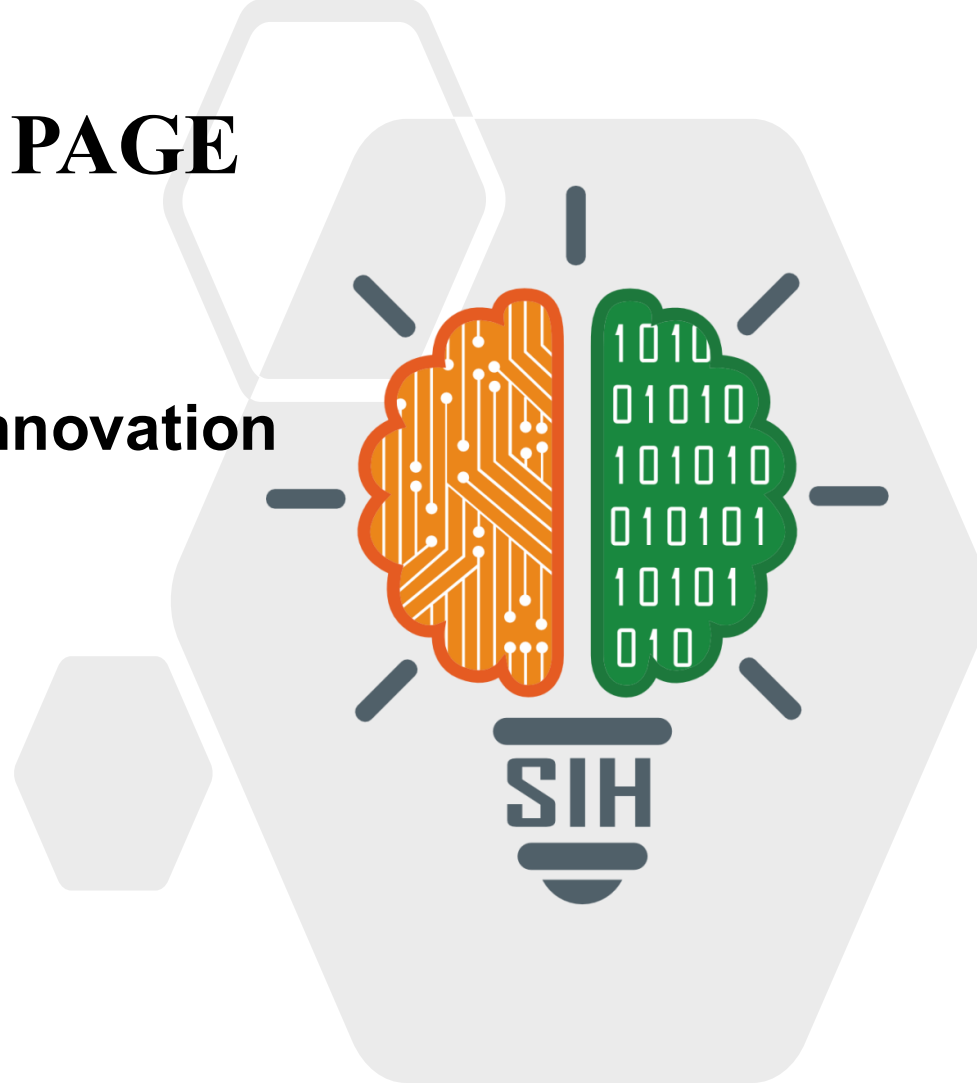


SMART INDIA HACKATHON 2024



TITLE PAGE

- **Problem Statement ID – 1533**
- **Problem Statement Title- Student innovation**
- **Theme- Robotics and Drones**
- **PS Category- Hardware**
- **Team ID-**
- **Team Name - Electro Elite**

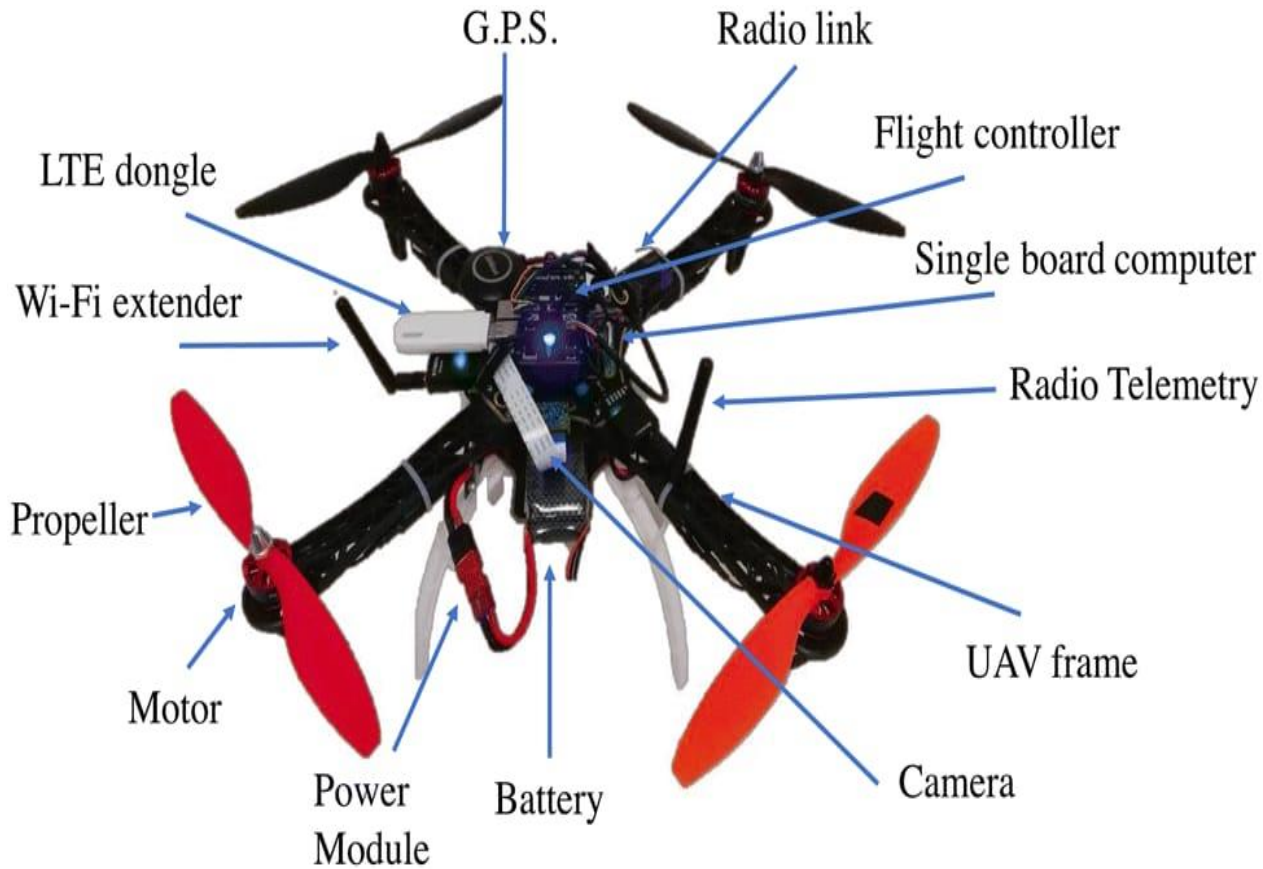


❖ Proposed Solution

- ❑ The Robots and drones involves in medical emergency and rescue operations with smart sensors, thermal cameras and self navigation systems to give quick response to emergencies by using embedded and IoT.
- ❑ Medical emergency response
- ❑ It includes handing over the deceased bodies to the appropriate authorities
- ❑ Provide first aid service to the trapped survivors
- ❑ Disaster response and recovery
- ❑ By listening to all sources of information, it can help the team to find the survivors more quickly

❖ **This is the solution's innovation and uniqueness:**

- ✓ Autonomous innovations - Drones can fly on their own and avoid obstacles
- ✓ Data collection - Drones collect data to improve future responses
- ✓ Enhanced situational Awareness - Drones provide a clear view of the situation



- ✓ Project Scope and Objectives
- ✓ System Architecture
- ✓ RF Communication System
- ✓ Flight Control System
- ✓ Power Management
- ✓ Integration and Testing
- ✓ Deployment and Maintenance
- ✓ Documentation and User Training

Feasibility:

1. High for tasks
2. Regulations may limit
3. Costs are decreasing, but specialized drones are still expensive
4. Improved batteries extend flight times

Viability:

1. High demand in agriculture, logistics etc
2. Easy to integrate
3. Privacy and airspace safety
4. Faster and more precise data collection
5. Wide range of applications



Impacts:

- Faster help arrives
- Help can reach remote areas
- Safer for responders and patients
- Better understanding of the situation
- Helps improve future responses

Benefits:

➤ Social:

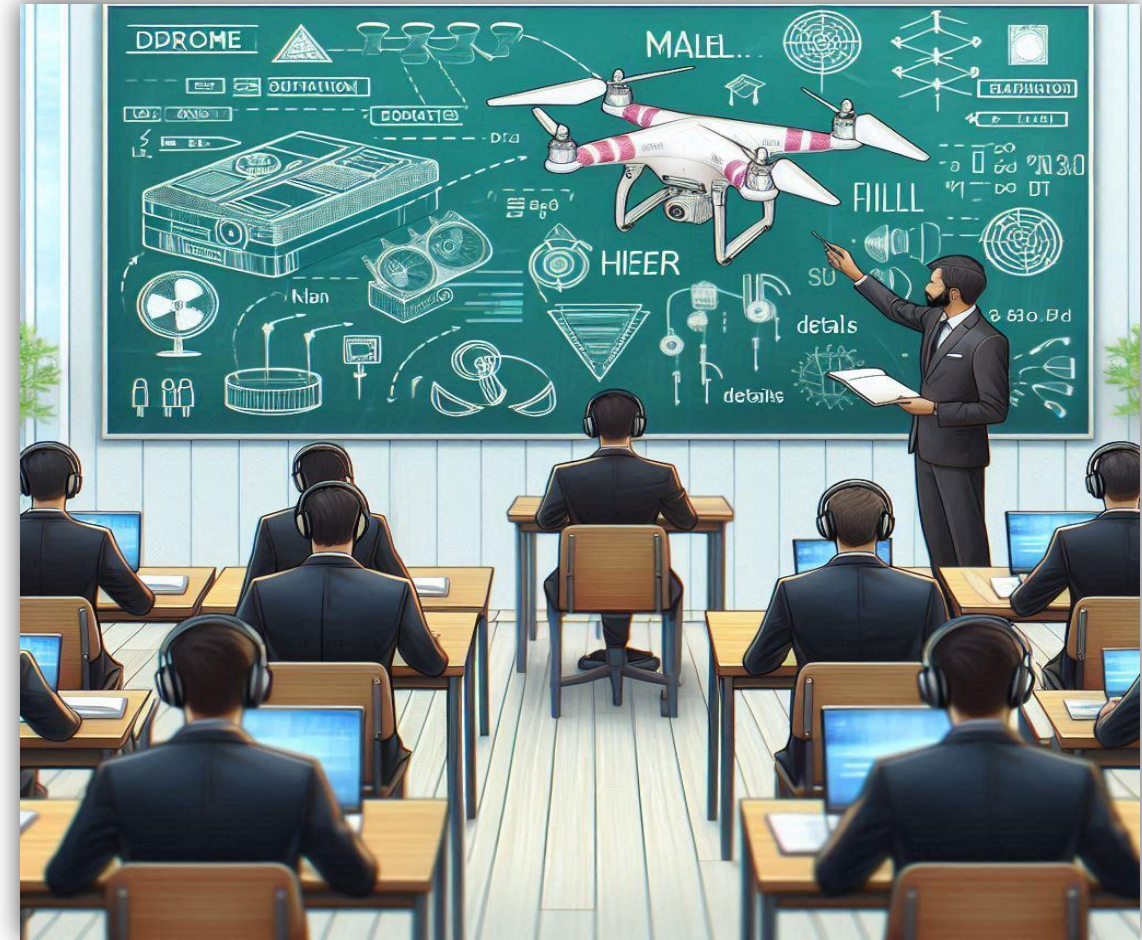
- ✓ Disaster management -Rapidly used in Earthquakes , Floods , wildfires

➤ Economic:

- ✓ Labor Cost - Using drones and robots can reduce the need for human workers in difficult or expensive tasks.

➤ Environmental:

- ✓ Drones help take care of
 - 1.solar panels
 - 2.Wind turbines



Research Papers & Articles:

☐ Medical drone delivery

- A Systematic Review" (2020)
- A Comprehensive Review" (2019)
- A Case Study" (2019)
- Regulatory Framework and Challenges" (2018)

☐ Search and Rescue Drones:

- "Drones Save Lives" by National Geographic (2019)
- "Robots to the Rescue" by Smithsonian Magazine (2020)
- "Search and Rescue Robotics" by CRC Press (2022)
- Wikipedia: "Search and Rescue Robotics" (2022)

Reference Link :

- <https://www.microdrones.com/en/drones/md4-300>
- <https://www.microdrones.com/en/drones/md4-3000/>
- <https://www.youtube.com/watch?v=bnoUBfLxZz0>